

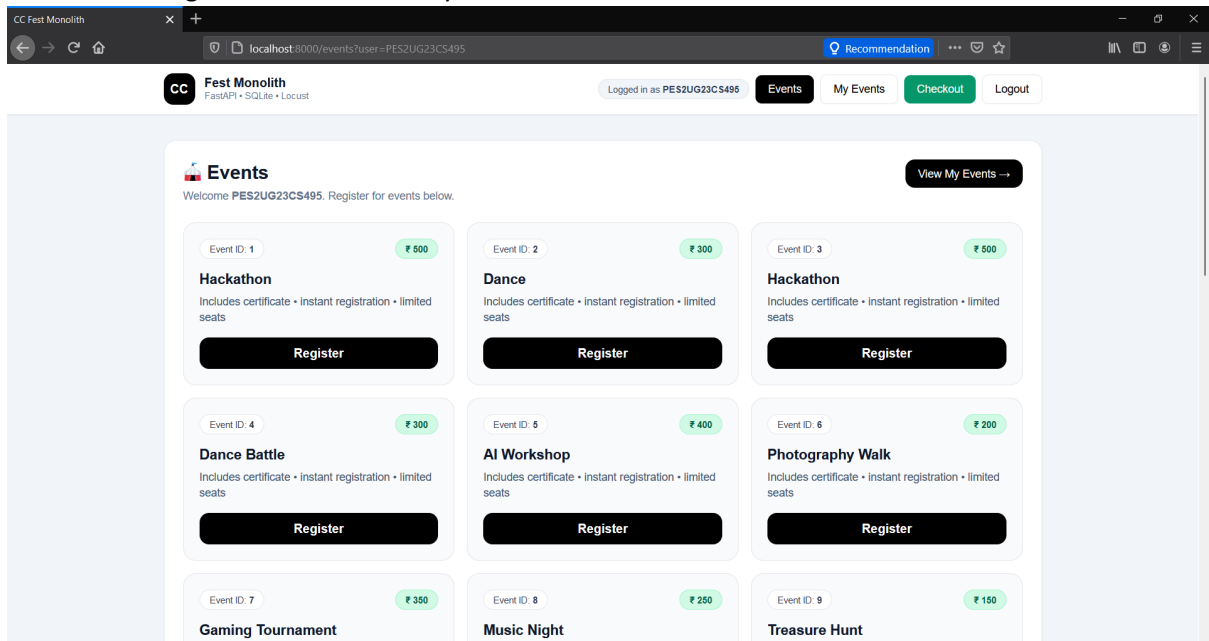
# UE23CS351B - CLOUD

## COMPUTING

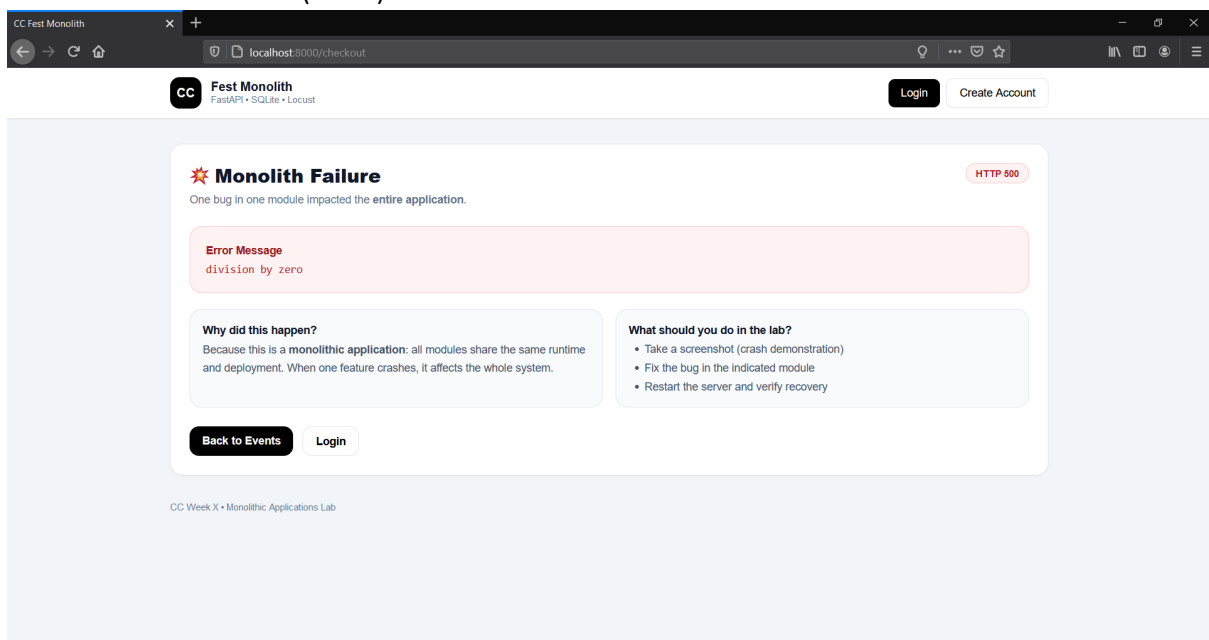
### LAB 2

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Date: 29/01/26

#### SS-1: Events Page Loaded Successfully



#### SS-2 : Monolithic Failure (Crash)



## SS-3 : Checkout Page Fixed

CC Fest Monolith

localhost:8000/checkout

FastAPI • SQLAle • Locust

LoginCreate Account

Checkout

This route is used to demonstrate a monolith crash + optimization.

Total Payable

₹ 6600

✓ After fixing + optimizing checkout logic, re-run Locust and compare results.

CC Week X • Monolithic Applications Lab

What you should observe

- One buggy feature can crash the entire monolith.
- Inefficient loops cause high response times under load.
- Optimization improves performance but architecture still scales as one unit.

Next Lab: Split this monolith into Microservices (Events / Registration / Checkout).

## SS-4 : Checkout Load Test (Before Optimization)

CC Fest Monolith

Locust

localhost:8009

67%

STATUS CLEANUP

RPS 0.7

Failures 0%

EDIT

STOP

RESET

STATISTICS

CHARTS

FAILURES

EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

LOGS

| Type       | Name      | # Requests | # Fails | Median (ms) | 95thile (ms) | 99thile (ms) | Average (ms) | Min (ms) | Max (ms) | Average size (bytes) | Current RPS | Current Failures/s |
|------------|-----------|------------|---------|-------------|--------------|--------------|--------------|----------|----------|----------------------|-------------|--------------------|
| GET        | /checkout | 19         | 0       | 5           | 2100         | 2100         | 112.66       | 4        | 2051     | 2797                 | 0.7         | 0                  |
| Aggregated |           | 19         | 0       | 5           | 2100         | 2100         | 112.66       | 4        | 2051     | 2797                 | 0.7         | 0                  |

```
(C:\Users\Surya\H\Documents\all programs\SQL_AleLab-2\FMSS3\SS4\locust> locust -f locust\checkout_locustfile.py
[2020-04-20 09:32:32.152] locust:locust.main: Starting Locust 2.4.3
[2020-04-20 09:32:32.152] locust:locust.main: Starting web interface at http://localhost:8009, press enter to open your default browser.
[2020-04-20 09:32:32.281] locust:locust.runner: Ramping to 1 users at a rate of 1.00 per second
[2020-04-20 09:32:32.281] locust:locust.runner: All users spawned. (Checkpoints: 1) (1 total users)
Feedback (most recent call last):
[2020-04-20 09:32:32.281] locust:locust.runner: www/lib/site-packages/gevent_futures.py, line 276, in python_check_callbacks
def python_check_callbacks(callback, watcher_obj): # pylint: disable=unused-argument

KeyboardInterrupt
[2020-04-20 09:34:04.889] locust:locust.main: Shutting down (exit code 0)
Type      Name      # reqs    # fails    Avg      Min      Max      Med     req/s    failures/s
-----
GET /checkout 19         0         5         2100     2100     112.66    4       2051     2797     0.7      0
Aggregated 19         0         5         2100     2100     112.66    4       2051     2797     0.7      0
```

## SS-5 : Checkout Load Test (After Optimization)

CC Fest Monolith

Locust

localhost:8009

67%

STATUS RUNNING

Users 1

RPS 0.6

Failures 0%

EDIT

STOP

RESET

STATISTICS

CHARTS

FAILURES

EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

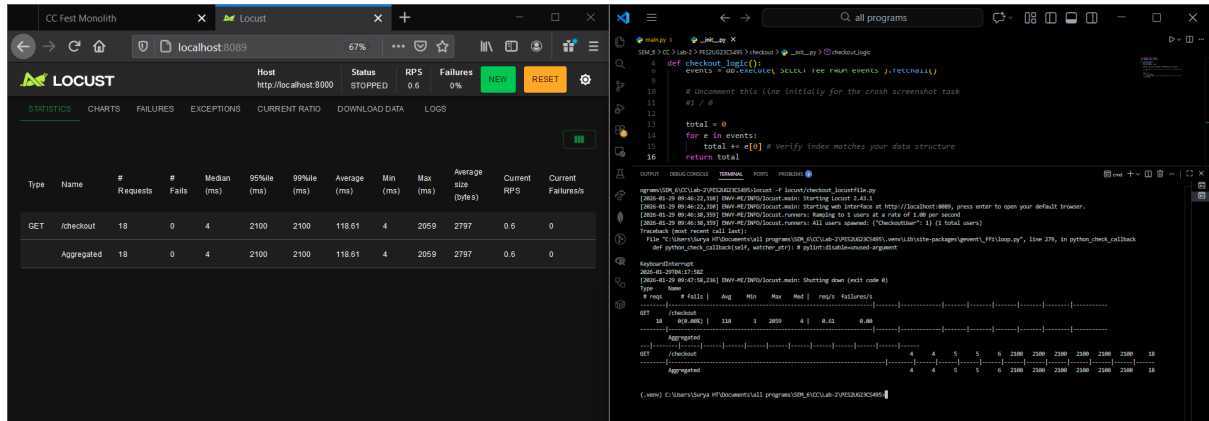
LOGS

| Type       | Name      | # Requests | # Fails | Median (ms) | 95thile (ms) | 99thile (ms) | Average (ms) | Min (ms) | Max (ms) | Average size (bytes) | Current RPS | Current Failures/s |
|------------|-----------|------------|---------|-------------|--------------|--------------|--------------|----------|----------|----------------------|-------------|--------------------|
| GET        | /checkout | 74         | 0       | 4           | 10           | 2000         | 33.03        | 4        | 2045     | 2797                 | 0.6         | 0                  |
| Aggregated |           | 74         | 0       | 4           | 10           | 2000         | 33.03        | 4        | 2045     | 2797                 | 0.6         | 0                  |

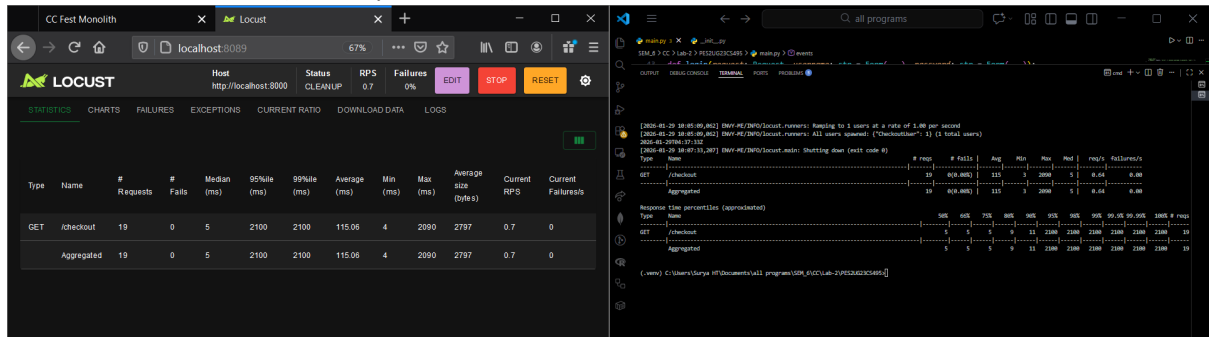
```
(C:\Users\Surya\H\Documents\all programs\SQL_AleLab-2\FMSS3\SS4\locust> locust -f locust\checkout_locustfile.py
[2020-04-20 09:38:22.807] locust:locust.main: Starting Locust 2.4.3
[2020-04-20 09:38:22.807] locust:locust.main: Starting web interface at http://localhost:8009, press enter to open your default browser.
[2020-04-20 09:38:36.481] locust:locust.runner: Ramping to 1 users at a rate of 1.00 per second
[2020-04-20 09:38:36.481] locust:locust.runner: All users spawned. (Checkpoints: 1) (1 total users)
Feedback (most recent call last):
[2020-04-20 09:38:36.481] locust:locust.runner: www/lib/site-packages/gevent_futures.py, line 276, in python_check_callbacks
def python_check_callbacks(callback, watcher_obj): # pylint: disable=unused-argument

KeyboardInterrupt
[2020-04-20 09:39:33.181] locust:locust.main: Shutting down (exit code 0)
Type      Name      # reqs    # fails    Avg      Min      Max      Med     req/s    failures/s
-----
GET /checkout 74         0         4         10       2000     33.03    4       2045     2797     0.6      0
Aggregated 74         0         4         10       2000     33.03    4       2045     2797     0.6      0
```

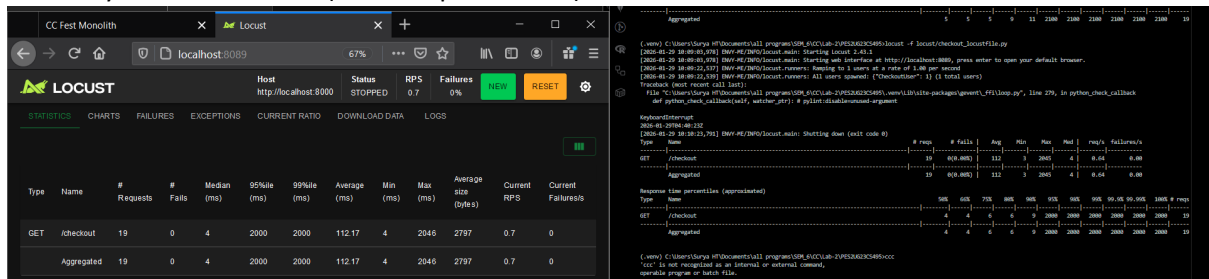
## SS-6 : Events Load Test (Before Optimization)



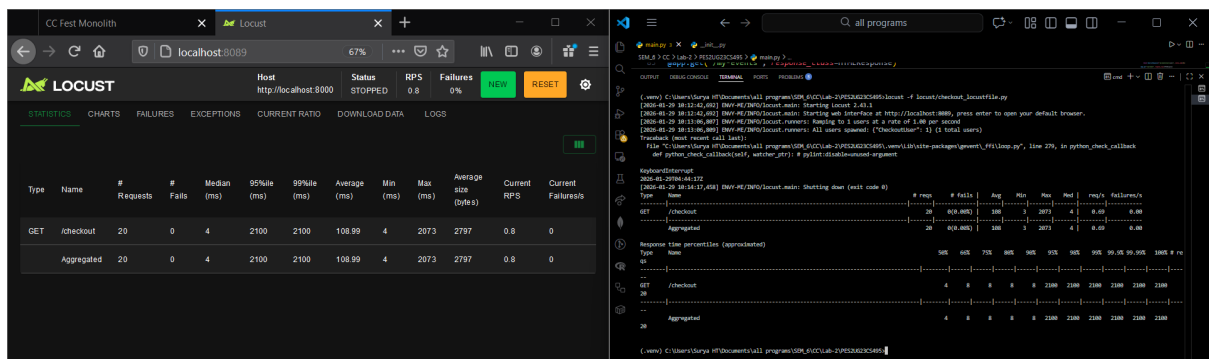
## SS-7 : Events Load Test (After Optimization)



## SS-8 : My Events Load Test (Before Optimization)



## SS-9 : My Events Load Test (After Optimization)



### 1. What was the bottleneck?

- **/checkout:** An inefficient while loop that incremented and decremented values one unit at a time, causing excessive CPU usage and  $O(N)$  time complexity based on the fee amount.
- **/events:** A busy-wait loop running three million iterations with a modulo operation, unnecessarily blocking the server.
- **/my-events:** A blocking loop running 1.5 million iterations with a dummy increment, adding artificial delay to each request.

### 2. What change did you make?

- **/checkout:** Replaced the iterative loop with a direct arithmetic calculation that sums event prices efficiently.
- **/events:** Removed the busy-wait loop and its related variable entirely.
- **/my-events:** Removed the dummy loop, leaving only the required SQL JOIN operation.

### 3. Why did the performance improve?

- **/checkout:** Reduced time complexity from being dependent on the fee value to being dependent only on the number of items, enabling instant computation.
- **/events:** Eliminated three million unnecessary CPU operations, significantly reducing response time.
- **/my-events:** Removed 1.5 million redundant iterations, allowing the endpoint to respond immediately and improving throughput (RPS).

**Github Link:** <https://github.com/RO-03/CC-Lab-2>